

Xingyu Wang

CONTACT INFORMATION	No. 1954 Huashan Road, Room 1508, Shanghai, China https://joshwang0322.github.io	+86 021 62933488 xingyu.wang@sjtu.edu.cn
RESEARCH INTERESTS	Rare Event Analysis, Large Deviations, Metastability, Heavy Tails; Machine Learning; Stochastic Simulation; Self-exciting Processes	
ACADEMIC POSITIONS	Antai College of Economics and Management, Shanghai Jiao Tong University Assistant Professor <i>Department of Management Science</i>	Shanghai, China 2026 - Present
	University of Amsterdam Postdoc Researcher <i>Quantitative Economics, Amsterdam School of Economics</i> <i>Hosts: Roger J.A. Laeven, Bert Zwart</i>	Amsterdam, Netherlands 2024 - 2026
EDUCATION	Northwestern University <i>Ph.D., Department of Industrial Engineering and Management Sciences</i> <i>Advisor: Chang-Han Rhee</i>	Evanston, IL, United States 2018 - 2024
	Northwestern University <i>M.S. in Analytics</i>	Evanston, IL, United States 2016 - 2017
	Peking University <i>B.S. in Psychology and Applied Mathematics</i>	Beijing, China 2011 - 2016
JOURNAL PAPERS ($\alpha - \beta$: ALPHABETICAL AUTHORSHIP)	[1] Sample Path Large Deviations for Multivariate Heavy-Tailed Hawkes Processes and Related Lévy Processes ($\alpha - \beta$) Jose Blanchet, Roger J. A. Laeven, Xingyu Wang, Bert Zwart <i>Bernoulli</i>, Accepted [2] Strongly Efficient Rare-Event Simulation for Regularly Varying Lévy Processes with Infinite Activities Xingyu Wang, Chang-Han Rhee <i>Mathematics of Operations Research</i>, Accepted	
SUBMITTED AND WORKING PAPERS	[3] Global Dynamics of Heavy-Tailed SGD's in Nonconvex Loss Landscape: Characterization and Control Xingyu Wang, Chang-Han Rhee Major Revision at <i>Operations Research</i>; arXiv:2510.20905 [4] Tail Asymptotics of Cluster Sizes in Multivariate Heavy-Tailed Hawkes Processes ($\alpha - \beta$) Jose Blanchet, Roger J. A. Laeven, Xingyu Wang, Bert Zwart Major Revision at <i>Annals of Applied Probability</i>; arXiv:2503.01004 [5] Large Deviations and Metastability Analysis for Heavy-Tailed Dynamical Systems Xingyu Wang, Chang-Han Rhee Minor Revision at <i>Annals of Applied Probability</i>; arXiv:2307.03479 <i>George Nicholson Student Paper Competition, Second Place</i>, 2023	

CONFERENCE
PAPERS

- [6] Multi-agent Multi-armed Bandit with Fully Heavy-tailed Dynamics
Xingyu Wang, Mengfan Xu
Winter Simulation Conference (WSC), 2025
- [7] Importance Sampling Strategy for Heavy-Tailed Systems with Catastrophe Principle
Xingyu Wang, Chang-Han Rhee
Winter Simulation Conference (WSC), 2023
- [8] Eliminating Sharp Minima from SGD with Truncated Heavy-Tailed Noises
Xingyu Wang, Sewoong Oh, Chang-Han Rhee
International Conference on Learning Representations (ICLR), 2022
Nemhauser Prize for Best Student Paper, 2022
- [9] Keyword-Based Topic Modeling and Keyword Selection
Xingyu Wang, Lida Zhang, Diego Klabjan
IEEE Big Data, 2021
- [10] Efficient Rare-Event Simulation for Multiple Jump Events in Regularly Varying Lévy Processes with Infinite Activities
Xingyu Wang, Chang-Han Rhee
Winter Simulation Conference (WSC), 2020
- [11] Competitive Multi-Agent Inverse Reinforcement Learning with Sub-Optimal Demonstrations
Xingyu Wang, Diego Klabjan
International Conference on Machine Learning (ICML), 2018

HONORS AND
AWARDS

- **Danish Data Science Academy Visit Grant,** 2026
- **Nemhauser Best Dissertation Prize,** Northwestern University 2025
- **George Nicholson Student Paper Competition, Second Place,** INFORMS 2023
- **Terminal Year Fellowship,** Northwestern University 2023
- **Nemhauser Prize for Best Student Paper,** Department of Industrial Engineering and Management Sciences, Northwestern University 2022
- **Benjamin A. Sachs Graduate Fellowship,** Northwestern University 2022
- **Arthur P. Hurter Award for Academic Excellence among First Year Graduate Students,** Department of Industrial Engineering and Management Sciences, Northwestern University 2019
- **Lee Wai Wang Scholarship,** Department of Psychology, Peking University 2015
- **National Scholarship for Undergraduates,** Department of Psychology, Peking University 2013

TALKS

- Large Deviations for Multivariate Heavy-Tailed Hawkes Processes**
- IISE Annual Conference & Expo, Arlington, TX May, 2026
 - INFORMS Annual Meeting, Atlanta, GA Oct, 2025
 - Seminar at School of Mathematics and Statistics, Beijing Institute of Technology Apr, 2025
 - Stochastic Seminars at Korteweg-de Vries Institute for Mathematics, Amsterdam, Netherlands Feb, 2025

Sharp Characterization and Control of Global Dynamics of SGD's with Heavy Tails

- University of Copenhagen, Statistics Seminar Sep, 2025
- Monte Carlo Methods 2025, Chicago, IL July, 2025
- Applied Probability Society Conference 2025, Atlanta, GA July, 2025
- Bayes Comp 2025, Singapore Jun, 2025
- Data-Driven Queueing Challenges Conference, Eindhoven, Netherlands Nov, 2024
- INI Satellite Programme on Heavy Tails in Machine Learning, London, United Kingdom Apr, 2024
- Cornell University, ORIE, Ithaca, NY Feb, 2024
- University of Pittsburgh, Industrial Engineering, Pittsburgh, PA Jan, 2024
- SNAPP Seminar, Lightning Talk Session, Virtual Dec, 2023

Importance Sampling Strategy for Heavy-Tailed Systems with Catastrophe Principle

- Winter Simulation Conference (Advanced Tutorial), San Antonio, TX Dec, 2023

Large Deviation and Metastability Analysis for Heavy-Tailed Dynamical Systems

- Neurips 2023, Heavy Tails in ML Workshop (Poster), New Orleans, LA Dec, 2023
- INFORMS Annual Meeting, Phoenix, AZ Oct, 2023
- Cornell ORIE Young Researchers Workshop, Ithaca, NY Oct, 2023

Eliminating Sharp Minima from SGD with Truncated Heavy-Tailed Noises

- INFORMS Annual Meeting, Phoenix, AZ Oct, 2023
- Applied Probability Society Conference 2023, Nancy, France Jun, 2023
- INFORMS Annual Meeting, Indianapolis, IN Oct, 2022
- International Conference on Learning Representations (2022), Virtual Apr, 2022
- DeepMath (2021), Virtual Nov, 2021

Efficient Rare-Event Simulation for Multiple Jump Events in Regularly Varying Lévy Processes with Infinite Activities

- Winter Simulation Conference, Virtual Dec, 2020
- INFORMS Annual Meeting, Virtual Nov, 2020

TEACHING

Theses Supervisor

- UvA, Master's Theses, Actuarial Science and Mathematical Finance 2025, 2026
- UvA, Bachelor's Theses, Actuarial Science 2025, 2026

Instructor

- IEMS Bootcamp (Probability) for PhD Students 2022 Fall, 2023 Fall

Teaching Assistant

- UvA Introduction to Data Science 2025

- UvA Non-Life Insurance: Statistical Techniques and Data Analytics 2024, 2025
- NU IEMS 317 Discrete Event Systems Simulation 2022 Winter; 2021 Winter
- NU IEMS 315 Stochastic Models 2021 Spring, Fall; 2020 Spring, Fall

Course Grader

- NU OPNS 450 Decision Models & Prescriptive Analytics 2022 Winter
- NU IEMS 435 Stochastic Simulation 2020 Winter

SERVICES

- Referee: *Operations Research, Management Science, Mathematics of Operations Research, INFORMS Journal on Computing, Journal of Machine Learning Research, Communications Physics, Computers and Operations Research, Journal of Simulation, ICML (2025), Winter Simulation Conference (2023, 2025)*
- Volunteer:
 - *NU MORE-REACH panel (2024)*
- Webmaster:
 - *Stochastic Networks, Applied Probability, and Performance (SNAPP) seminar (2023 - 2024)*
- Mentor:
 - *1st-year mentorship program at NU INFORMS Student Chapter (2020, 2021)*

Last Update: October 6, 2026